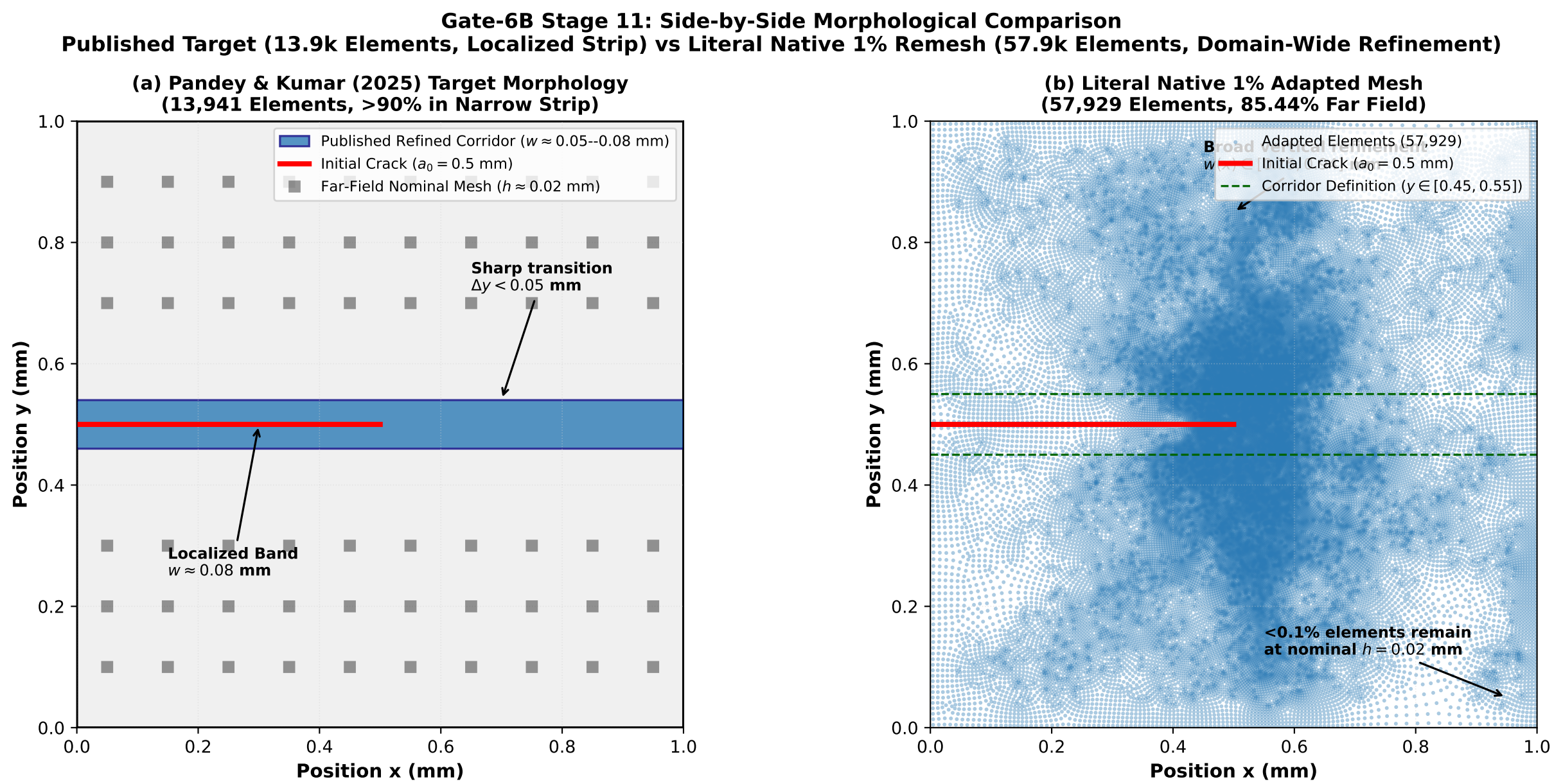

Gate-6B Stage 11: Side-by-Side Morphological Comparison
Published Target (13.9k Elements, Localized Strip) vs Literal Native 1% Remesh (57.9k Elements, Domain-Wide Refinement)
(a) Pandey & Kumar (2025) Target Morphology
(13,941 Elements, >90% in Narrow Strip)
Published Refined Corridor ($w \approx 0.05$--0.08 mm)
Initial Crack ($a_0 = 0.5$ mm)
Far-Field Nominal Mesh ($h \approx 0.02$ mm)
Sharp transition
$\Delta y < 0.05$ mm
Localized Band
$w \approx 0.08$ mm
Position y (mm)
Position x (mm)
(b) Literal Native 1% Adapted Mesh
(57,929 Elements, 85.44% Far Field)
Adapted Elements (57,929)
Initial Crack ($a_0 = 0.5$ mm)
Corridor Definition ($y \in [0.45, 0.55]$)
<0.1% elements remain
at nominal $h = 0.02$ mm
Position y (mm)
Position x (mm)